

## ADVANCED REVIEW



WILEY

# A survey on federated learning in data mining

Bin Yu<sup>1</sup> | Wenjie Mao<sup>1</sup> | Yihan Lv<sup>1</sup> | Chen Zhang<sup>1</sup> | Yu Xie<sup>2</sup>

<sup>1</sup>School of Computer Science and Technology, Xidian University, Xi'an, China

<sup>2</sup>Key Laboratory of Computational Intelligence and Chinese Information Processing of Ministry of Education, Shanxi University, Taiyuan, China

## Correspondence

Yu Xie, Key Laboratory of Computational Intelligence and Chinese Information Processing of Ministry of Education, Shanxi University, Taiyuan, China.  
Email: sxlljcx@gmail.com

## Funding information

National Key Research and Development Program of China, Grant/Award Numbers: 2020YFC1523304, 2019YFC1606705; National Natural Science Foundation of China, Grant/Award Number: 62106131; Key Research and Development Program of Shaanxi Province, Grant/Award Number: 2019ZDLGY17-01

**Edited by:** Witold Pedrycz, Editor-in-Chief

## Abstract

Data mining is a process to extract unknown, hidden, and potentially useful information from data. But the problem of data island makes it arduous for people to collect and analyze scattered data, and there is also a privacy security issue when mining data. A collaboratively decentralized approach called federated learning unites multiple participants to generate a shareable global optimal model and keeps privacy-sensitive data on local devices, which may bring great hope to us for solving the problems of decentralized data and privacy protection. Though federated learning has been widely used, few systematic studies have been conducted on the subject of federated learning in data mining. Hence, different from prior reviews in this field, we make a comprehensive summary and provide a novel taxonomy of the application of federated learning in data mining. This article starts by providing a thorough description of the relevant definitions and concepts, followed by an in-depth investigation on the challenges faced by federated learning. In this context, we elaborate four taxonomies of major applications of federated learning in data mining, including education, healthcare, IoT, and intelligent transportation, and discuss them comprehensively. Finally, we discuss four promising research directions for further research, that is, privacy enhancement, improvement of communication efficiency, heterogeneous system processing, and reducing economic costs.

This article is categorized under:

Technologies > Machine Learning

Fundamental Concepts of Data and Knowledge > Big Data Mining

Technologies > Artificial Intelligence

## KEYWORDS

collaboratively decentralized method, data mining, federated learning

## 1 | INTRODUCTION

With the increasing popularity of computing devices, such as wearable devices, smartphones, and autonomous vehicles, people are generating a wealth of valuable data each day (T. Li, et al., 2020; Zhao et al., 2018). Data mining uses machine learning, statistics, and database methods to extract potentially valuable information from these massive amounts of data (Vaghashia & Ganatra, 2015), and is widely used in various domains, such as medical care, finance, and data science, to provide people with better services (Ganesh & Christy, 2015). In order to analyze and use data

efficiently and conveniently, data mining researchers tend to collect data from many scattered devices and store them in a central location, such as a database or virtual data warehouse. However, this method faces two problems. One is that the collection and centralized storage of such data is time-consuming and expensive (Lyu et al., 2020) due to the lack of interconnection between different data owners, resulting in the widespread occurrence of data islands. Another concern is with the increasingly strict requirements of the public for privacy protection (C. Zhang, Xie, et al., 2021), the practice of data aggregation is becoming less feasible (T. Li et al., 2020). Especially when the training dataset of machine learning applications contains sensitive information, it will pose a serious security threat to user data (Mammen, 2021). For example, although services such as embedded positioning technology have been widely used, the owners of these devices are reluctant to share their data directly, because this data may contain sensitive information (Abadi et al., 2016), such as home addresses, medical records, and financial privacy. It is likely to infringe on privacy intentionally or unintentionally when data mining methods are utilized directly for comprehensive data analysis. Accordingly, how to extract valuable information from decentralized data islands while preserving private data is a challenging problem.

In recent years, privacy-preserving data mining was proposed, which aims to mine the required information from massive data and protect sensitive information without sacrificing the utility of the data (Vaghashia & Ganatra, 2015). The majority of the existing research on privacy-preserving data mining can be divided into two aspects (X. Qi & Zong, 2012). One is to preserve sensitive data in the process of data mining. The other is to preserve the private results produced by the application of data mining.

However, no privacy-preserving algorithm is superior to all other algorithms in all possible standards (Vaghashia & Ganatra, 2015). Actually, in terms of protecting data privacy, the key point of the problem is how to securely obtain results rather than data. On the other hand, in order to protect privacy, the transmission of data on different devices is bound to be restricted, and the establishment of a centralized data warehouse is also difficult to implement. Furthermore, methods of storing data on devices and sharing models are becoming increasingly attractive from privacy, security, regulatory and economic perspectives (Zhao et al., 2018). In this case, introducing federated learning into data mining becomes a better choice.

Federated learning (Konečný et al., 2016) is a collaborative learning method tailored for isolated clients, in which data is retained locally without centralized collection, and model parameters from local models are shared and aggregated. It helps address the problem of insufficient samples and combines decentralized participants with data. In addition, the small update of the centralized model is transmitted during the training process and provides stronger privacy security than the traditional machine-learning method of collecting and storing data in a central location. Thus, federated learning is very beneficial in resolving privacy issues associated with data islands. In general, federated learning can be applied to almost any edge device, which may completely change key fields such as medical treatment, transportation, finance, and smart homes (Mammen, 2021). However, there are also several challenges associated with federated learning. The challenges can be classified into four categories: data-related problems, communication-related challenges, device-related problems, and privacy security challenges. Data-related problems include the heterogeneity of data, complex data processing, and unbalanced data distribution. In practice, it is unrealistic to assume that the local data on each edge device is always IID. Whereas communication encompasses unreliable connections and low efficiency. Many studies have been done to reduce the matrix weight factor in the transmission. Security challenges include privacy leaks, poisoning attacks, and server-side attacks.

But these challenges do not affect federated learning as a promising method. Based on this, this article discusses federated learning in data mining in-depth.

## 1.1 | Purpose of study

The purpose of this research is to review the work of federated learning in the field of data mining comprehensively. Although many researchers have applied federated learning to quite a few areas of data mining, their works are fragmented or have not interpreted these studies from the perspective of data mining. In addition, as an emerging field, federated learning still has many problems and challenges in the actual application process. Based on this, our work tries to answer the following questions:

- Why does data mining need federated learning?
- Which fields related to data mining can use federated learning to gain higher benefits?

- When it comes to federated learning in data mining, what challenges are there? What are the future research directions?

## 1.2 | Contributions

Currently, there are many reviews on data mining and federated learning, but none of these studies has been done on federated learning in data mining. The work contributes to a comprehensive overview of federated learning in data mining in terms of definition, challenges, and applicability. Several works have studied federated learning, but what distinguishes our work from previous works is that we manage to delve deeply into the applications of federated learning in data mining. This work provides promising ideas for data mining practitioners to apply federated learning in lots of fields and alleviate data island challenges. Overall, the contribution of this article lies in the following aspects:

- We present a brief description of the definitions, principles, and processes of federated learning in data mining.
- We highlight the major applications of federated learning to illustrate the advantages of introducing federated learning into data mining.
- We bring to light the key challenges of federated learning in data mining and put forward possible research directions in the future.

## 1.3 | Paper organization

The this article is organized as follows. We sketch the overview of data mining and federated learning in Section 2. Section 3 summarizes four major challenges in the application of federated learning. Then, Section 4 goes over applications in the four hot fields of data mining. In Section 5, according to the current application of federated learning in data mining, we discuss how to deal with the emerging challenges, and propose four promising research directions for the future. In the end, Section 6 concludes the article.

## 2 | PRELIMINARIES

In this section, we underline in detail the concepts and terminology involved in the survey, and the topics are data mining and federated learning.

### 2.1 | Data mining

Data mining aims to extract useful patterns from huge amounts of data (Fayyad et al., 1996; Gupta, 2014; Shah & Nair, 2015) and transform it into an understandable structure for future use. The five steps of knowledge discovery process, one of the most important research topics (Tomar & Agarwal, 2013), are selection, preprocessing, transformation, data mining, and evaluation. In the data mining phase, several machine learning algorithms are used to generate knowledge or patterns from data. The patterns are conveyed in decision trees, clusters, or association rules (Vaghashia & Ganatra, 2015), which help data holders analyze the relationship between data and make decisions. As a means of identifying patterns from massive data, data mining has attracted the attention of many researchers. However, this knowledge may contain sensitive information, which seriously threatens the security of private information (Ciriani et al., 2008). Therefore, privacy protection in data mining is a key issue. In this subsection, we will discuss privacy-preserving data mining, which is summarized as follows.

Privacy-preserving data mining techniques: The research on privacy protection methods focuses on data distortion, data encryption, data publishing (X. Qi & Zong, 2012), etc., such as privacy protection classification mining algorithms (Weiping et al., 2006), privacy protection association rule mining (Luo et al., 2005; P. Zhang et al., 2006), distributed privacy protection collaborative recommendation (Hui-You, 2006), and so on.

- **Data Distortion Based Technique:** A method for distorting sensitive data while keeping data or data attributes unchanged. Through the application of certain interference methods, such as randomization, adding noise, aggregation, the original data is transformed and distorted, to prevent attackers from obtaining the original data. However, it is required to ensure that the processed data can still maintain statistical properties for data mining and other operations.
- **Cryptography Based Technique:** The technology applies encryption to hide sensitive data in the process of data mining, which is often used in distributed environments, such as secure multi-party computing (Clifton et al., 2002; A. C.-C. Yao, 1986). Secure multi-party computation involves independent multi-party participants working together to generate output results based on private inputs, and it is one of the most widely used privacy-preserving methods. Multiparty cooperation ensures that the input information is secure, and no party may obtain private data from another.
- **Restricted Publishing Technique:** This technique conditionally publishes data according to specific conditions (Sweeney, 2002a, 2002b). The  $k$ -anonymity algorithm establishes  $k - 1$  records for each record that are indistinguishable from any record of published data information. In general, the larger the value of  $k$ , the better the privacy-preserving effect of the anonymous model. (Ciriani et al. (2008) described the concept of  $k$ -anonymity, illustrated different implementation methods, and introduced possible methods to ensure  $k$ -anonymity in data mining.
- **Reconstruction Based Technique:** The original distribution of data is reassembled from randomized data by applying reconstruction-based algorithms. Many proposed privacy-preserving data mining techniques use data perturbation or reconstruction in the data convergence layer. Chang and Moskowitz (1998) studied the construction of decision tree classifiers using disturbed single record values as training data and shows that when data miners know the background by reconstructing the distribution, the estimation of privacy will be reduced. Agrawal and Srikant (2000) improved the Bayesian reconstruction process by using the EM algorithm on distributed data.

To sum up, the technology based on data distortion has high efficiency, but there is a certain degree of information loss. The technology based on encryption is just the opposite, which can ensure the accuracy and security of the final data, but the computational overhead is relatively large. And the advantage of restricted publishing technology is that it can ensure that the published data is true, but the published data will lose partial information. Reconstruction based technique is a relatively simple method of obscuring personal information, but it is unsuitable for multi-attribute databases.

## 2.2 | Federated learning

Federated learning is a learning paradigm that aims to address the problem of data islands while protecting data privacy (L. Li, Fan, & Lin, 2020). Different from aggregating the source data to the server, federated learning is based on decentralized participants as well as the training process, putting the learning process on edge devices (Mothukuri et al., 2021). The use of federated learning is widespread in transportation, healthcare, IoT, security systems, and mobile applications (Aledhari et al., 2020). Enormous potential of federated learning has attracted the attention of researchers (Yu et al., 2020a; Zhuo et al., 2019). Federated learning trains models directly on end-user devices or isolated data centers and keeps data localized, while aggregating only temporarily updated parameters on a centralized server. On the other hand, federated learning can generate more robust models without sharing data across devices, resulting in privacy-preserving solutions with higher security, making it especially appropriate for privacy-sensitive environments (M. Chen et al., 2019; Lu et al., 2019b).

The strict regulations regarding data privacy make it impractical for customers' information to be gathered and shared in a central location. Therefore, traditional machine learning algorithms are hampered because they usually need a large quantity of data for training (Larson et al., 2020) and a master server to deal with data storage and training models. However, whether building a data pipeline so that data can flow through a server, or transferring a machine learning model to any device, it's not optimal because their models do not converge quickly (Aledhari et al., 2020). In federated learning, the model is distributed to decentralized devices for training as well as prediction (T. Li et al., 2019), and the parameters are brought over to the master server for aggregation. In such a scenario, it is also advantageous to apply the machine learning process to low-cost devices, such as mobile phones and PDA (Doku et al., 2019).

The steps of federated learning are as follows. Under the coordination of the central server, clients (such as laptops, smartphones, tablet PCs, etc.) cooperate to train the model. In each of a series of rounds, the server selects a random subset of clients to participate in training. Each selected client downloads copies of the global model parameters to

update local models with local data. Then, clients upload parameters after training, and the server accumulates them into the new global model to average the contribution. Next, the central server dispatches the averaged update to clients (L. Li, Fan, & Lin, 2020). After several rounds of iteration, a model that is extremely close to the traditional centralized machine learning model is generated for achieving the expected performance. Figure 1 shows the general architecture of federated learning. Even in its most basic form, federated learning exhibits two significant privacy advantages over uploading training data to the server:

1. It transmits smaller increments of model parameters required for model training, and the update usually contains much less information.
2. Model-updating is a short-lived transfer and merging of data into the global model.

Next, we briefly introduce the federated averaging algorithm with formulas. For quintessential federated learning, thousands of clients train a global model  $w$  and upload gradients to the server. The objective function of the central server is

$$\min_w F(w), \text{ where } F(w) = \sum_{k=1}^m \frac{n_k}{n} F_k(w) \quad (1)$$

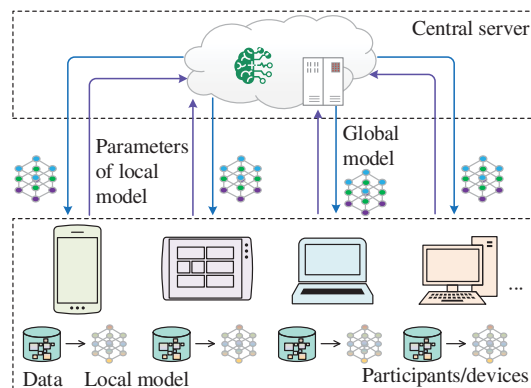
where  $m$  and  $n$  denote the total number of clients participating in the training and the total data of all clients, respectively. The  $k$ th client holds  $n_k$  data samples with  $n_k = |D_k|$ .  $D_k$  is local dataset of the  $k$ th client assigned from the whole data set  $D$ , with  $D = \cup_{k=1}^m D_k$  (McMahan et al., 2017).  $F_k(w)$  represents the local objective function of the  $k$ th device, which is defined as:

$$F_k(w) = \frac{1}{n_k} \sum_{i \in D_k} f_i(w) \quad (2)$$

Federated learning maps the input value  $x_i$  to the output tag  $y_i$  for predicting unknown data and the size of input-output pairs  $(x_i, y_i)$  is  $n$ . In Equation (2),  $f_i(w) = f(x_i, y_i, w)$  is the loss function generated by the model with parameter  $w$  for pair  $(x_i, y_i)$  in dataset  $D_k$ . The sum of the loss functions generated by all instances in  $D_k$  divided by the total quantity of data generated by the client  $k$  is the average loss function. The goal of the model is to minimize  $f_i(w)$  in the training process (Abdul Rahman et al., 2021), so that the model can better predict the  $i$ th data sample.

### 3 | CHALLENGES IN FEDERATED LEARNING

Federated learning has drawn people's attention since it was proposed. Although lots of studies about it have been conducted, there are many challenges for application, which may be caused by objective problems in the training process or attacks launched by malicious adversaries. In this section, we classify the challenges into four categories and discuss



**FIGURE 1** A schematic architecture of a general federated learning system

them separately. Furthermore, we provide a brief overview of two scalability issues, that is, system disruption and unavailability.

### 3.1 | Data problems

#### 3.1.1 | Non-IID data

In probability and statistics, non-IID means that in a dataset, the random variables have a different distribution, so the probability distribution for one client and another in federated learning is  $p_i \neq p_j$ . However, unique device types and usages will affect the generation of data. Similarly, the data on different devices may be non-identically distributed due to each client corresponding to a particular situation (such as user, geographic location, and time window; Kairouz et al., 2019), which will affect the performance of federated learning model (T. Li et al., 2020; McMahan et al., 2017). The traditional distributed framework algorithm only works well with IID data, but with non-IID data, it will lead to problems such as difficulty in convergence and occurrence of too many communication rounds. The research works in Bonawitz et al. (2019) and Y. Liu, James, et al. (2020) attempted to solve this problem by modifying the method of model aggregation. Duan et al. (2020) proposed a federated learning framework FedGroup based on a similarity clustering strategy, which solves the statistical heterogeneity by dividing the federated optimization into subgroups and realizing the approximate IID distribution in the group. Moreover, the method proposed by Zhao et al. (2018) works by creating a small dataset that is globally shared among all edge devices to improve the training effectiveness.

#### 3.1.2 | Complex data processing

Existing studies have shown that federated learning can be applied in many domains. However, there are still obstacles in fully integrating federated learning into other environments, especially in terms of data. To ensure that the model is as accurate as possible, large sets of data are required to make the model and algorithm learn effectively to produce the best outcomes (Zhu et al., 2020). Unfortunately, data record forms may be completely different across multiple horizontals. In addition, the data sizes in different clients may vary greatly, resulting in uneven distribution. In order to solve this problem, the general solution is to add some extra processes in the data preprocessing stage. It is not suitable for large-scale data training because of the high cost of parameter conversion (L. Li, Fan, Tse, et al., 2020). Moreover, even handling the data itself can be challenging due to the diversity of contents, file formats, and data structures.

#### 3.1.3 | Massively distributed and unbalanced data

Many reasons lead to significant differences in the amount of training data, such as different usage of devices, the local environment of clients, and noninteraction between clients. In federated learning, millions of clients may participate in this process, ranging from mobile phones to IoT devices, vehicles, and so on (Abdul Rahman et al., 2021).

### 3.2 | Communication problems

#### 3.2.1 | Unreliable connection

The network connections status varies from client to client. In most cases, connections of clients are typically limited, expensive, unavailable and slow, which limits the number of connections available in the process of federated learning. Unreliable network connection may yield inefficient results in training the global model, because many of the accessible clients may suffer from high network delay and dropout at any time. Decentralized training (Xie et al., 2020) is proven to be more effective than centralized training on networks with low bandwidth or high latency in federated learning.

### 3.2.2 | Communication overhead

One of the key challenges has been the communication overhead in the process of federated learning (Mammen, 2021). Modern data is characterized by a short life cycle, fast iterative updates, and large amount. Reducing communication overhead is crucial for federated learning to deal with large amounts of data flexibly (L. Li, Fan, Tse, et al., 2020). In federated learning, model updating requires continuous communication between server and multiple clients to update model parameters, which will be a serious burden on the network. The time of processing data has gradually decreased due to improvements in computing power, and low communication transmission efficiency of federated learning has become the main bottleneck, limiting its training speed (Konečný et al., 2016). Existing research attempts to address the problem of communication overheads by model compression or reducing the amount of data sent back to the central server by the client (Hsieh et al., 2017; P. P. Liang et al., 2020; Luping et al., 2019; Malekijoo et al., 2021) or by LAN-WAN orchestration (Yuan et al., 2020). Yurochkin et al. (2019) designed a Bayesian nonparametric federated learning framework, which can aggregate local models into a federated model without extra parameters, thus reducing communication rounds.

## 3.3 | System heterogeneity

### 3.3.1 | Heterogeneous hardware

Each device in a federated learning framework may have different computing and communication capabilities due to varying hardware and network connections. In general, only a small proportion of devices are active simultaneously on the network. It is also possible that a lot of devices are unreliable, so heterogeneity exacerbates the challenge of fault tolerance. Stripelis and Ambite (2021) introduced a new semi-synchronous federated learning protocol, which periodically mixes local models and performances well in heterogeneous data and computing environments. And T. Li et al. (2018) presented a framework FedProx to solve the heterogeneity problem in federated network. Moreover, Laguel et al. (2020) proposed a federated learning framework  $\Delta$  – FL to filter out heterogeneous client devices that do not conform to the population data distribution.

### 3.3.2 | Poor device performance

There are differences in performance among the devices participating in the training process, especially the IoT devices, whose physically available memory and storage space are limited. The devices may exit abnormally during continuous high-intensity training and calculation. Hence, the uneven performance of the equipment participating in federated learning is a serious concern. Nishio and Yonetani (2019) introduced a framework FedCS, which selects appropriate clients for aggregation in the process of federated learning.

## 3.4 | Security threats

Although the private data on clients never comes out of local storage during training process, the federated system is not safe enough, because the transmission of parameters may lead to indirect privacy leakage (Bhowmick et al., 2018; Bos et al., 2014; Kairouz et al., 2019). Some researchers consider retrieving data in the federated learning framework. General attack types are mainly divided into the following three categories:

### 3.4.1 | Privacy leak

In federated learning, models may be prone to attacks in real-world applications, because the motivation of clients participating in training is difficult to assess, and the credibility of the central server is difficult to verify. Protecting user privacy only by model-updating is obviously inadequate. There have been several studies demonstrating that gradient information can leak sensitive private information. Based on the gradient information uploaded by the client, an

adversary can indirectly determine the label information and the member information of the data set. Carlini et al. (2018) extracted the user's sensitive data from the recurrent neural network (RNN). Geiping et al. (2020) proved that the feasibility of reconstructing input data from gradient information is not determined by the deep network architecture, and they recovered a batch of input images using cosine similarity and counter-attack methods. Ma et al. (2020) proposed a method to protect privacy under the federated learning framework, which achieves a better privacy protection effect by increasing the number of tests on the server and the number of epochs on each client.

### 3.4.2 | Poisoning attacks

In this article, poisoning attacks are divided into data poisoning, model poisoning, and backdoor.

- **Data Poisoning Attacks:** Adversaries may take advantage of the anonymity of the client to participate in federated learning and they need many malicious participants since they have no direct access to raw data (Bagdasaryan et al., 2020). Malicious attackers can directly inject toxic data into the target device or embed disguised data to destroy data integrity and indirectly affect the target node (G. Sun et al., 2020). This kind of attack can be targeted or nontargeted. The targeted poisoning attack reduces the accuracy of the global model and affects the prediction of other categories by targeted means (Tolpegin et al., 2020). By deliberately poisoning the training data to participate in the training of the local model, the attacker may eventually give rise to model misclassification. This type of attack can be mitigated by identifying malicious participants according to the model parameters uploaded by each client before averaging the model in each round of learning.
- **Model Poisoning Attacks:** Federated learning is generally vulnerable to model poisoning attacks because it cannot ensure that participants are not malicious. Model poisoning attempts to poison the local model to damage the global model, such as by changing the parameters of local models to launch attacks so as to achieve a more powerful attack than training data poisoning (Bagdasaryan et al., 2020). We can reject the poisoned model by detecting the error rate and loss function (Mammen, 2021). Using data poisoning prevention measures or anomaly detection requires access to participants' training data or model updates, which will undermine participants' privacy. In addition, the federation learned to adopt the method of secure aggregation, which can prevent anyone from reviewing participants' data or updates, but the aggregator is not able to detect anomalies. The work conducted by Bhagoji et al. (2019) shows that Byzantine-resilient aggregation is weak to defense this attack. Further, J. Zhang et al. (2019) attempted to use generative adversarial networks to generate training data for model poisoning attacks.
- **Backdoor Attacks:** As for backdoor attacks, the federated model can be backdoored by one or multiple malicious participants using model replacement (Bagdasaryan et al., 2020). It's possible for the backdoor to label certain tasks incorrectly while retaining the accuracy of the global model. Z. Sun et al. (2019) studied how to implement backdoor attack. Bagdasaryan et al. (2020) introduced backdoor into the global model and proved that any client participating in the federated learning steps can present the hidden backdoor function in the shared global model. The defense measure can be weak differential privacy or the norm threshold of updates (Mammen, 2021). In addition, model pruning and fine-tuning are suggested as solutions to mitigate the risk of backdoor attacks (Jiang et al., 2019). The backdoor attack is triggered only when the model gets specific input, and then leads to the wrong output of the neural network, so it is exceedingly hidden and time-consuming to identify backdoor attacks. Moreover, a new distributed backdoor attack shows that attackers may divide a trigger into multiple parts and embedded each part into different nodes, which bring new form to security evaluation (L. Li, Fan, Tse, et al., 2020).

### 3.4.3 | Server-side attacks

Malicious servers can easily inspect private data from clients and alter the global model to build malicious tasks. Z. Wang, Song, et al. (2019) considered attacks from server in the process of federated learning. They used multitask generative adversarial networks to distinguish users' identities by attacking the privacy of clients. Furthermore, there is a risk that the original data will be cracked by back deduction. By attacking the central server, malicious adversaries

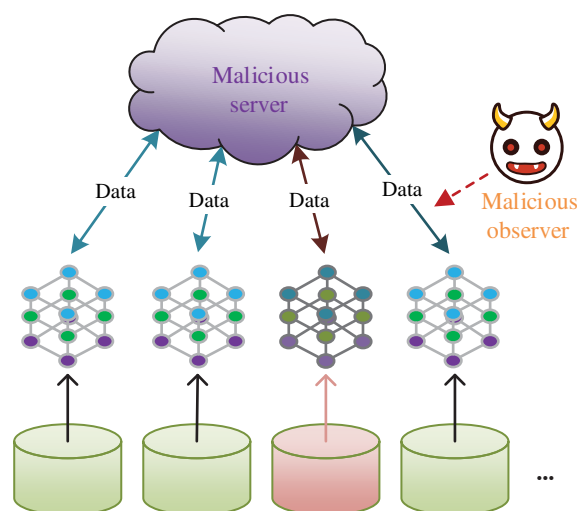
attempt to infer information from the updates of models in the training process (Mammen, 2021). The defense measure against this member reasoning attack is to find a mechanism to provide privacy protection.

To summarize, existing privacy-preserving algorithms cannot effectively protect users' privacy. As shown in Melis et al. (2019), an attacker may infer a specific location of the data from the training data of clients. Since participants can freely join and exit a round during the whole training process, the federated learning setting will lead to the emergence of various threat models and vulnerabilities of many participants (Abdul Rahman et al., 2021). There are three major types of threats to federated learning as shown in Figure 2 and are classified as follows:

1. Malicious clients modify local updates to carry out poisoning attacks.
2. Malicious observer infers the privacy information of the source data through the analysis of the model updates between trusted clients and the server.
3. Malicious server attempts to inspect the updates from clients and tamper the model.

Overall, the general privacy-preserving techniques are mainly divided into three categories, that is, secure computation, differential privacy, and a trusted execution environment.

- Security computation includes two technologies: secure multiparty computation and homomorphic encryption. For the former, in addition to the calculation results, each participant cannot obtain any input information from other entities and does not disclose the output value to other members participating in the calculation. Bonawitz et al. (2016) used secure multiparty computation to develop federated learning systems. Whereas the calculation by the latter is based on encrypted input. Cheng et al. (2021) proposed a lossless tree-boosting system named SecureBoost for privacy protection. SecureBoost is shown to be more accurate than nonfederated tree-boosting systems while maintaining data privacy.
- Differential privacy masked the clients' contribution by adding noise to clipped model parameters before aggregation (Bhowmick et al., 2018; Geyer et al., 2017). However, model accuracy may be compromised. Geyer et al. (2017) introduced an algorithm to address data leakage using differential privacy, which includes two methods: random subsampling and distorting aggregated update. N. Agarwal et al. (2018) proposed a method that takes into account both communication efficiency and differential privacy, and applied model compression to federated learning, which further improved the performance of federated learning and privacy protection. The results of Choudhury et al. (2019) show that the performance of federated learning is close to that of centralized system without differential privacy.
- Compared to traditional protection methods such as homomorphic encryption, the trusted execution environment provides lower overhead and greater privacy (Mo & Haddadi, 2019). Nevertheless, the complexity of hardware makes this a hard solution to scale in edge computing.



**FIGURE 2** Three types security threats in federated learning

### 3.5 | Other scalability issues

#### 3.5.1 | System disruption

Production system downtime is an inevitable threat to the IT industry (Mothukuri et al., 2021). People may experience backend server downtime, which may be due to some planned or unexpected situations. Since there is a local global model on each client node, the severity of this threat is low in federated learning, and the training process can be recovered after shutdown. But this is also a considerable threat, because it may be an attack well planned by the attacker to steal information from the federated learning environment.

#### 3.5.2 | Unavailability

During the training process, customer unavailability may produce inefficient results when training the global model. In this case, due to hardware problems or any other unexpected roadblocks, the client inadvertently missed the opportunity to participate in the training process. However, the probability is lower, and the aggregation algorithms working asynchronously can be selected to alleviate the threat (Mothukuri et al., 2021).

In this section, we highlight several challenges for the practical application of federated learning, among which communication-related and privacy security problems are the key challenges. Efforts on federated learning security and privacy have proposed lossless and effective methods without reducing the original accuracy (Abdul Rahman et al., 2021). However, some of these methods either incorporate extra hyperparameters or cause additional communication overheads, which will affect the accuracy and the communication. New technologies that limit the power of potential attackers can provide security and privacy guarantees and improve performance in situations where security and privacy are essential.

## 4 | APPLICATIONS OF FEDERATED LEARNING IN DATA MINING

Federated learning has been applied in many different fields. In light of the amount of existing research and its applicability in a particular field, we select the four most active research fields and deeply discuss the application of federated learning in data mining: education, healthcare, the IoT, and smart transportation, which will be discussed in this section.

### 4.1 | Education

Educational data mining has been a popular application of data mining in the past 10 years. It aims to collect users' learning behavior data and use knowledge in computer science, data mining, and other fields under the guidance of psychology and learning science to discover how students learn. Educational data mining relies on data obtained from schools, students, and administrators, such as test scores and attendance records. These data are all stored in the computer information systems of their respective schools or educational institutions, forming independent islands of information. How to obtain data from different educational backgrounds or domains is a topic of concern. S. Guo and Zeng (2020) proposed FEEDAN, an educational data analysis framework based on federated learning. By using this framework, many institutions can form pedagogical data federations. They only share analytical capabilities instead of directly exchanging their teaching data, so that they keep the data in their own place to guarantee their students' privacy invariably.

With the emergence of intelligent education systems and large-scale online open courses on the Internet, students' personalized education has received more attention. Personalized education can customize the most suitable learning plan for different students, significantly improving their learning efficiency. The key to realizing personalized education is knowledge tracing. The aim of knowledge tracing is to predict how students will perform in future assessments based on their knowledge over time (Piech et al., 2015). The earliest models of knowledge tracking used first-order Markov models, such as Bayesian knowledge tracing. In 2015, the method of deep learning was introduced into knowledge tracing, which achieved remarkable success and was widely used in many educational applications. However, deep

knowledge tracing still faces challenges because the data island problem also exists in the field of education. In order to solve the challenges of data scarcity, data imbalance, and difficult data comparison caused by data isolation, J. Wu et al. (2021) introduced the federated learning framework into deep knowledge tracing to train deep knowledge tracing models of each educational island to improve the quality of the model. And their study is the first work in this field.

## 4.2 | Healthcare

In the medical field, electronic health records have become an crucial source of real-world healthcare data for predicting disease incidence, patient response to treatment, and other events (Glicksberg et al., 2018). Through the analysis of electronic health records, data mining can predict diseases and provide personalized patient care. Electronic health records have been processed using traditional machine learning mechanisms, but there is a false assumption that electronic health records can easily be stored and shared in a centralized location (Aledhari et al., 2020). In fact, medical data comes from patients in various medical institutions and clinics, and its nature is sensitive. For protecting the privacy and security of patient data, federated learning has been introduced into the medical field in recent years. The authors of the literature J. Lee et al. (2018) aimed to develop a learning algorithm that allows learning hash codes to find similar patients from different hospitals without disclosing patient-level information. They introduced federated learning to ensure that this patient hash framework can protect patients' privacy. Unlike federated learning based on differential privacy, a syntactic anonymization method that offers privacy in federated learning is presented in Choudhury et al. (2019). In order to prove that the solution can effectively protect privacy and provide interpretability, two large-scale health data sets are used to evaluate models.

Moreover, because federated learning supports predictive modeling from different sources, the introduction of federated learning into the work of data mining can provide clinicians with more experience in the treatment of different diseases, and the medical systems can also benefit from it (Boughorbel et al., 2019). The federated learning model is leveraged in Brisimi et al. (2018) to predict future hospitalization for cardiac events, based on their medical history as described in the electronic health records. Boughorbel et al. (2019) incorporated uncertainty modeling into the federated learning algorithm model to predict preterm birth from distributed electronic health records. And this algorithm is simple while having good performance.

After the outbreak of COVID-19 in 2019, countless researchers worked hard to solve COVID-19. Federated learning, as a promising technology, has also been introduced into the new crown of medical research. Vaid et al. (2020) used federated learning to predict mortality within 7 days of hospitalized new COVID-19 patients. The patient data comes from electronic health records scattered in different medical institutions. W. Zhang, Zhou, et al. (2021) designed an architecture for dynamic fusion-based federated learning systems to improve accuracy and communication efficiency while preserving data privacy for COVID-19 detection.

## 4.3 | Internet of things

The rapid development of IoT provides ubiquitous sensing and computing capabilities (Nguyen et al., 2021), enabling massive amounts of data to be generated and connected to the internet from data creation devices such as smart phones, transportation facilities, public services, and household appliances. Through data mining, a large amount of data generated in IoT can be efficiently transmitted, analyzed, and processed (Shadroo & Rahmani, 2018). However, most data mining technologies still rely on the centralized architecture of the data center, such as cloud servers, which means that local data must be uploaded to the cloud, resulting in security and privacy problems. Federated learning provides more attractive functions to help data mining process data from different devices on the IoT more efficiently and safely. In the industrial internet of things, the study in Lu et al. (2019a) designed a data-sharing architecture authorized by blockchain, in which blockchain enables secure data retrieval and accurate model training. This process effectively maintains data privacy through a shared data model. Fu et al. (2020) proposed a privacy-protecting and verifiable federated learning framework to achieve efficient and secure model training for industrial scenes sensitive to training sets.

In addition to security concerns, storing large-scale distributed and private data sets in the cloud also faces challenges such as data heterogeneity and large economic expenditures. The breakthrough of edge computing technologies means that many controls in IoT do not need to be handed over to the cloud, but are completed at the local edge computing layer (J. Liu et al., 2021). This avoid the communication load and economic expenditure of uploading data to the

center while reducing the load on the cloud and increasing the speed of data processing. At the same time, the federated learning framework has been successfully used in edge computing systems, and the use of edge and terminal computing can meet the requirements of network edge for cloud capacity and equipment (Zheng et al., 2021). Q. Wu et al. (2020) advocated the deployment of a personalized federated learning framework in the cloud edge architecture to protect the privacy of IoT users and alleviate the impact of heterogeneity. In this framework, edge computing technology is used to achieve high communication and reduce computing costs. The authors in X. Wang, Han, et al. (2019) designed the “in-edge AI” framework, which integrates deep reinforcement learning techniques and a federated learning framework with mobile edge systems. They reduce unnecessary system communication load while optimizing mobile edge computing and caching.

As a result, some IoT applications require short response times (such as industrial applications, emergency management, real-time systems) and rely on resource-constrained equipment. The use of traditional data mining technology is neither effective nor efficient (Savaglio et al., 2019). Xu et al. (2021) proposed an efficient federated learning framework to solve the problem of deep neural networks being infeasible or extremely slow due to limited computing resources and poor network connections. It consists of three stages of pipelines: structured pruning, weight quantization, and selective updating. They jointly reduce the cost of computing, storage, and communication, so as to accelerate the training process of federated learning.

#### 4.4 | Smart transportation

The smart transportation system generates a huge amount of traffic information, such as road data, monitoring base stations, vehicle types, and traffic volume. Data mining can discover hidden value from massive amounts of information, and provide traffic forecasting, vehicle control, and other services, such as predicting public transportation delays, taxi demand analysis, and so on (Balbin et al., 2020; H. Yao et al., 2018). Due to privacy restrictions, most data mining research is based on public traffic data sets, and cannot comprehensively study traffic conditions. The use of federated learning technology can provide more comprehensive and accurate services for the intelligent transportation system. As shown in Y. Qi et al. (2021), federated learning is leveraged to protect the privacy of vehicle data in the task of traffic flow prediction. Additionally, their work utilizes blockchain technology to prevent malicious attackers from poisoning data. Y. Liu, James, et al. (2020) proposed a gated recurrent unit neural network algorithm based on federated learning to predict traffic flow. In order to achieve higher accuracy and scalability in the task of traffic flow prediction, they improve the federated average algorithm and develop a scheme based on ensemble clustering. The authors of X. Liang et al. (2019) designed a federated learning framework to enhance reinforcement transfer learning tasks in autonomous driving. They conduct experiments on remote-controlled cars and simulators respectively to verify the effectiveness of federated learning in the real collision avoidance system.

Another area worth pouring attention into in smart transportation is the internet of vehicles. The internet of vehicles is a new component of smart transportation. The vehicle-mounted sensor network provides a large amount of sensor information, which can directly affect intelligent traffic control and can detect vehicles through vehicle-to-infrastructure communication, thereby realizing intelligent transportation in a cost-effective and efficient way. The introduction of horizontal federated learning contributes to a more powerful machine learning model. In addition to using federated learning to train models locally, communication between vehicles is also crucial on the internet. A distributed, federated learning method has recently been considered in Samarakoon et al. (2019) to improve the reliability of vehicular communication while reducing network delay. Lyapunov optimize the communication network for vehicle users and achieve significant benefits in terms of power consumption. The work in Uprety et al. (2021) protects the data security on the internet vehicles. The authors propose an internet of vehicles' privacy protection misbehavior detection system using federated learning to train models for preventing data falsification attacks in vehicular networks.

For clarity, we list the taxonomy of the applications of federated learning in data mining in Table 1.

### 5 | FUTURE DIRECTIONS

As an active and continuous research field, the work of federated learning in data mining is in its infancy, and there are still many key open issues need to be explored. In this section, we propose four research directions worthy of discussion.

**TABLE 1** Taxonomy of federated learning application areas

| Applications         | Authors                       | Key contributions                                                                                            | Advantages                                                                                              | Disadvantages                                                                            |
|----------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Education            | S. Guo and Zeng (2020)        | Propose a federated learning based pedagogical data analysis framework.                                      | Build the federated learning server locally without changing the existing information system.           | Efficiency of model has not been considered.                                             |
|                      | J. Wu et al. (2021)           | Proposed a federated deep knowledge tracking framework.                                                      | Data can be expanded to alleviate data scarcity.                                                        | Students' privacy is not considered.                                                     |
| Healthcare           | J. Lee et al. (2018)          | Propose a privacy-preserving platform in a federated setting.                                                | Independently predict the incidence rate of five diseases and no risk of privacy leakage.               | Based on the assumption that the site has common feature events.                         |
|                      | Choudhury et al. (2019)       | Propose the syntactic anonymization approach for offering privacy in federated learning.                     | More interpretable at the level of privacy it offers.                                                   | Communication efficiency has not been investigated.                                      |
|                      | Brisimi et al. (2018)         | Develop a general decentralized optimization framework to predict hospitalizations.                          | Improve convergence rate and reduce communication cost.                                                 | Lack of simulation results achieved by federated learning.                               |
|                      | Boughorbel et al. (2019)      | Propose a premature delivery prediction learning algorithm based on federated uncertainty perception.        | Algorithm is simple and has good performance.                                                           | Do not design a rejection criterion for model.                                           |
|                      | Vaid et al. (2020)            | Propose a federated learning approach to predict the mortality of inpatients.                                | Reflect the effectiveness and practicality of federated learning.                                       | Accessibility and expanding analysis of federated models have not been considered.       |
|                      | W. Zhang, Zhou, et al. (2021) | Propose a dynamic fusion-based federated learning approach for medical diagnostic image analysis.            | Perform good in accuracy and communication efficiency.                                                  | Privacy issues has not been considered.                                                  |
| Internet of things   | Lu et al. (2019a)             | Propose a privacy-preserving data sharing architecture authorized by blockchain.                             | The solution achieves high efficiency and utility.                                                      | The application of blockchain technology to protect data privacy is still an open issue. |
|                      | Fu et al. (2020)              | Propose a privacy-preserving and verifiable federated learning framework for industrial intelligent.         | The computational overhead of verification mechanism does not increase with the number of participants. | The label classification of the research data is not rich enough.                        |
|                      | Q. Wu et al. (2020)           | Advocate a personalized federated learning framework in a cloud-edge architecture.                           | Suitable for large-scale deployment, have high accuracy and low communication cost.                     | Efficiency of model has not been considered.                                             |
|                      | X. Wang et al. (2019)         | Propose the "In-Edge AI" framework for a better deployment of intelligent resource management.               | The scheme has advantages in the balance of performance and cost.                                       | Learning computing task needs to be further optimized.                                   |
|                      | Xu et al. (2021)              | Propose a federated learning framework to accelerate the model training on resource-constrained IoT devices. | Control the training overhead while still guaranteeing the learning performance.                        | No-IID data has not been considered.                                                     |
| Smart transportation | Y. Qi et al. (2021)           | Propose a federated learning framework for protecting privacy in traffic flow prediction.                    | Prevent participants from a membership inference attack.                                                | Communication overhead for federated learning is expensive.                              |

(Continues)

TABLE 1 (Continued)

| Applications | Authors                      | Key contributions                                                                  | Advantages                                                                 | Disadvantages                                                         |
|--------------|------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
|              | Y. Liu, James, et al. (2020) | Design a federated learning algorithm for predicting traffic flow.                 | Help to reduce communication overhead large-scale models.                  | Numerical simulations for privacy is lacked.                          |
|              | X. Liang et al. (2019)       | Propose a federated learning scheme for traffic collision avoidance tasks.         | Can transfer reinforcement learning agents knowledge in trail-time.        | Latency for federated learning simulations has not been investigated. |
|              | Samarakoon et al. (2019)     | Propose a federated learning approach for optimizing vehicular communications.     | Improve communication quality and reduce latency.                          | No-IID data has not been considered.                                  |
|              | Upreti et al. (2021)         | Propose a federated learning approach for protecting privacy in vehicular network. | The model converged with acceptable accuracy, precision, and recall value. | The weights sent over the network are not encrypted.                  |

## 5.1 | Privacy enhancement

The protection of privacy is an overwhelmingly critical issue in federated learning (Aledhari et al., 2020). Although one of the purposes of federated learning is to protect privacy, today's privacy protection scheme is still facing challenges. Leakage of private information in areas with strict regulations and government policies such as medical data is unacceptable. In order to protect private data, many solutions implement strict privacy restrictions. Unfortunately, these good privacy protection schemes are implemented at the expense of accuracy (T. Li et al., 2020). A few researchers attempt to give solutions, such as the work in Huang et al. (2020) recently used neural network pruning to replace the method of adding differentially private noise to neural networks, which protects privacy while not losing much data utility. In addition to balancing efficiency and privacy limits, another problem is that different fields have different provisions for data privacy, so there is no one knife-cut solution. This requires the researchers of federated learning to make efforts to provide personalized privacy protection.

## 5.2 | Improvement of communication efficiency

Communication between the server and the client is essential. The server and remote clients need to communicate continuously to exchange model update information. As the number of participating clients increases, it will bring a huge bandwidth burden to the communication network, and the communication efficiency of federated learning becomes the main factor limiting its training speed. The model needs to be designed in a way that is efficient for communication (F. Chen et al., 2018), which means that we cannot send the entire training data set for the training process. A better way is to send smaller messages or model updates every time. In that case, two factors need to be taken into consideration. One is to reduce the number of communication rounds and the other is to reduce the size of each message in each round.

## 5.3 | Heterogeneous system processing

In the federated learning environment, the performance of the equipment participating in the training varies greatly. For example, there may be differences in hardware configuration, network bandwidth, and battery capacity between clients. And different terminal devices have different computing capabilities, communication speeds, and storage capabilities (T. Li et al., 2020). Besides, because of the instability of the participants, the number of participants in the learning process may not be fixed. Especially in training involving millions of devices, only hundreds of clients are active, so each client is not necessarily reliable. Many current studies have not considered the addition of new parties or the departure of existing parties. There are several studies that attempt to solve this problem. For example, Google's system (Bonawitz et al., 2019) can tolerate device disconnection, but researchers need to do more in this direction.

## 5.4 | Economic cost reduction

How to establish an incentive mechanism to make all data owners be honest and voluntarily join the data federation at anytime and anywhere is a major challenge. If there is not much economic benefit to compensate for the consumption of computing resources, users may be unwilling to use their resources for data training. The current ideal construction is to reward organizations that contribute more data by rewarding them more, and implementing incentives through a consensus mechanism. The current academic circles mainly provide trust and incentive mechanisms for federated learning by combining blockchain technology. By using a block-chained federated learning architecture, the work in Kim et al. (2018) can leverage the blockchain to exchange and verify local learning model updates for mobile devices. The associated miners in the blockchain network are responsible for recording model updates and providing corresponding rewards.

## 6 | CONCLUSION

Federated learning is a collaboratively machine learning paradigm that can mitigate many of the systemic privacy risks and handle the problem of data island, which are also challenges faced by data mining. This study shows that applying federated learning to data mining can better promote the development of data mining and solve the issues. Extensive research on either federated learning or data mining has been conducted in the past few years, but there is no survey conducted on federated learning in data mining. In this paper, we provide a comprehensive survey of the topic. First, we present a detailed overview of the basic definition, process, as well as algorithms of data mining and federated learning. Then, we highlight four major challenges existing in the practical application of federated learning, which are data-related problems, communication-related challenges, device-related problems, and privacy security challenges, among which communication-related and privacy security problems are the key challenges, followed by a brief description of two scalability issues. Then, based on thorough analysis of the main technical challenges and current related work, the four most active domains of federated learning in data mining are discussed in detail, including education, healthcare, IoT, and intelligent transportation. Additionally, we also discuss their key issues, advantages, and disadvantages, which have been briefly summarized in Table 1. Finally, we point out future directions in four aspects: privacy enhancement, improvement of communication efficiency, heterogeneous system processing, and reducing economic costs. This will help researchers make better use of federated learning for data mining and open the door for future research in this field.

## CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

## AUTHOR CONTRIBUTIONS

**Bin Yu:** Project administration (equal); resources (equal); supervision (equal). **Wenjie Mao:** Conceptualization (equal); formal analysis (equal); investigation (equal); methodology (equal); visualization (equal). **Yihan Lv:** Conceptualization (equal); formal analysis (equal); investigation (equal); methodology (equal). **Chen Zhang:** Funding acquisition (equal); methodology (equal); project administration (equal); resources (equal); supervision (equal). **Yu Xie:** Formal analysis (equal); funding acquisition (equal); investigation (equal); methodology (equal); project administration (equal); resources (equal); supervision (equal).

## DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

## ORCID

Yu Xie  <https://orcid.org/0000-0002-3431-0432>

## RELATED WIREs ARTICLES

[Big data analytics-A review of data-mining models for small and medium enterprises in the transportation sector](#)

## FURTHER READING

Agarwal, S. (2013). Data mining: Data mining concepts and techniques. In 2013 International Conference on Machine Intelligence and Research Advancement. Katra, India. pp. 203–207.

- Alom, M. Z., Taha, T. M., Yakopcic, C., Westberg, S., Sidike, P., Nasrin, M. S., Esesn, B. C. V., Awwa, A. A. S., & Asari, V. K. (2018). The history began from alexnet: A comprehensive survey on deep learning approaches. *arXiv preprint arXiv:1803.01164*.
- Alsrehin, N. O., Klaib, A. F., & Magableh, A. (2019). Intelligent transportation and control systems using data mining and machine learning techniques: A comprehensive study. *IEEE Access*, 7, 49830–49857.
- Bonawitz, K., Ivanov, V., Kreuter, B., Marcedone, A., McMahan, H. B., Patel, S., Ramage, D., Segal, A., & Seth, K. (2017). Practical secure aggregation for privacy-preserving machine learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*. Dallas Texas, USA. pp. 1175–1191.
- Elbir, A. M., Soner, B., & Coleri, S. (2020). Federated learning in vehicular networks. *arXiv preprint arXiv:2006.01412*.
- Guo, H., Liu, A., & Lau, V. K. (2020). Analog gradient aggregation for federated learning over wireless networks: Customized design and convergence analysis. *IEEE Internet of Things Journal*, 8(1), 197–210.
- Han, J., Pei, J., & Kamber, M. (2011). *Data mining: Concepts and techniques*. Elsevier.
- Huang, Y. (2021). F1bench: A benchmark suite for federated learning. In *Intelligent computing and block chain: First Benchcouncil International Federated Conferences, FICC 2020, Qingdao, China, October 30-november 3, 2020: Revised selected papers*. Vol. 1385, p. 166.
- Jia, B., Zhang, X., Liu, J., Zhang, Y., Huang, K., & Liang, Y. (2021). Blockchain-enabled federated learning data protection aggregation scheme with differential privacy and homomorphic encryption in IIoT. *IEEE Transactions on Industrial Informatics*, p. 1
- Jin, S., Luo, X., & Ma, D. (2018). Determining the breakpoints of fundamental diagrams. *IEEE Intelligent Transportation Systems Magazine*, 12(1), 74–90.
- Khan, L. U., Pandey, S. R., Tran, N. H., Saad, W., Han, Z., Nguyen, M. N., & Hong, C. S. (2020). Federated learning for edge networks: Resource optimization and incentive mechanism. *IEEE Communications Magazine*, 58(10), 88–93.
- Lee, G. H., & Shin, S.-Y. (2020). Federated learning on clinical benchmark data: Performance assessment. *Journal of Medical Internet Research*, 22(10), e20891.
- Lim, W. Y. B., Huang, J., Xiong, Z., Kang, J., Niyato, D., Hua, X.-S., Leung, C., & Miao, C. (2021). Towards federated learning in UAV-enabled internet of vehicles: A multi-dimensional contract-matching approach. *IEEE Transactions on Intelligent Transportation Systems*, 22(8), 5140–5154.
- Liu, Y., Yuan, X., Xiong, Z., Kang, J., Wang, X., & Niyato, D. (2020). Federated learning for 6g communications: Challenges, methods, and future directions. *China Communications*, 17(9), 105–118.
- Pfohl, S. R., Dai, A. M., & Heller, K. (2019). Federated and differentially private learning for electronic health records. *arXiv preprint arXiv:1911.05861*.
- Sharma, P., Shamout, F. E., & Clifton, D. A. (2019). cc. *arXiv preprint arXiv:1912.00354*.
- Sheller, M. J., Edwards, B., Reina, G. A., Martin, J., Pati, S., Kotrotsou, A., Milchenko, M., Xu, W., Marcus, D., Colen, R. R., & Bakas, S. (2020). Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10(1), 1–12.
- SÜZEN, A. A., & ŞİMŞEK, M. A. (2020). A novel approach to machine learning application to protection privacy data in healthcare: Federated learning. *Namik Kemal Tıp Dergisi*, 8(1), 22–30.
- Truex, S., Baracaldo, N., Anwar, A., Steinke, T., Ludwig, H., Zhang, R., & Zhou, Y. (2019). A hybrid approach to privacy-preserving federated learning. In *Proceedings of the 12th ACM Workshop on Artificial Intelligence and Security*. London, UK. pp. 1–11.
- Wahab, O. A., Mourad, A., Otrok, H., & Taleb, T. (2021). Federated machine learning: Survey, multi-level classification, desirable criteria and future directions in communication and networking systems. *IEEE Communications Surveys & Tutorials*, 23(2), 1342–1397.
- Weizhong, Q., Deqing, Z., & Hai, J. (2007). Research on privacy preservation mechanism for credentials and policies in grid computing environment. *Journal of Computer Research and Development*, 44(1), 11.
- Yedder, H. B., Cardoen, B., & Hamarneh, G. (2021). Deep learning for biomedical image reconstruction: A survey. *Artificial Intelligence Review*, 54(1), 215–251.
- Yu, H., Liu, Z., Liu, Y., Chen, T., Cong, M., Weng, X., Niyato, D., & Yang, Q. (2020b). A sustainable incentive scheme for federated learning. *IEEE Intelligent Systems*, 35(4), 58–69.
- Zhan, Y., Li, P., Qu, Z., Zeng, D., & Guo, S. (2020). A learning-based incentive mechanism for federated learning. *IEEE Internet of Things Journal*, 7(7), 6360–6368.

## REFERENCES

- Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. Vienna, Austria. pp. 308–318.
- Abdul Rahman, S., Tout, H., Ould-Slimane, H., Mourad, A., Talhi, C., & Guizani, M. (2021). A survey on federated learning: The journey from centralized to distributed onsite learning and beyond. *IEEE Internet of Things Journal*, 8(7), 5476–5497.
- Agarwal, N., Suresh, A. T., Yu, F., Kumar, S., & McMahan, H. B. (2018). cpSGD: Communication-efficient and differentially-private distributed SGD. *arXiv preprint arXiv:1805.10559*.
- Agrawal, R., & Srikant, R. (2000). Privacy-preserving data mining. In *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*. Dallas, TX. pp. 439–450.
- Aledhari, M., Razzak, R., Parizi, R. M., & Saeed, F. (2020). Federated learning: A survey on enabling technologies, protocols, and applications. *IEEE Access*, 8, 140699–140725.

- Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., & Shmatikov, V. (2020). How to backdoor federated learning. In International conference on artificial intelligence and statistics. Palermo, Italy. pp. 2938–2948.
- Balbin, P. P. F., Barker, J. C., Leung, C. K., Tran, M., Wall, R. P., & Cuzzocrea, A. (2020). Predictive analytics on open big data for supporting smart transportation services. *Procedia Computer Science*, 176, 3009–3018.
- Bhagoji, A. N., Chakraborty, S., Mittal, P., & Calo, S. (2019). Analyzing federated learning through an adversarial lens. In International conference on machine learning. Long Beach, CA. pp. 634–643.
- Bhowmick, A., Duchi, J., Freudiger, J., Kapoor, G., & Rogers, R. (2018). Protection against reconstruction and its applications in private federated learning. *arXiv preprint arXiv:1812.00984*.
- Bonawitz, K., Eichner, H., Grieskamp, W., Huba, D., Ingerman, A., Ivanov, V., Kiddon, C., Konečný, J., Konečný, S., McMahan, H. B., Overveldt, T. V., Petrou, D., Ramage, D., & Roselander, J. (2019). Towards federated learning at scale: System design. *arXiv preprint arXiv:1902.01046*.
- Bonawitz, K., Ivanov, V., Kreuter, B., Marcedone, A., McMahan, H. B., Patel, S., Ramage, D., Segal, A., & Seth, K. (2016). Practical secure aggregation for federated learning on user-held data. *arXiv preprint arXiv:1611.04482*.
- Bos, J. W., Lauter, K., & Naehrig, M. (2014). Private predictive analysis on encrypted medical data. *Journal of Biomedical Informatics*, 50, 234–243.
- Boughorbel, S., Jarray, F., Venugopal, N., Moosa, S., Elhadi, H., & Makhlof, M. (2019). Federated uncertainty-aware learning for distributed hospital EHR data. *arXiv preprint arXiv:1910.12191*.
- Brisimi, T. S., Chen, R., Mela, T., Olshevsky, A., Paschalidis, I. C., & Shi, W. (2018). Federated learning of predictive models from federated electronic health records. *International Journal of Medical Informatics*, 112, 59–67.
- Carlini, N., Liu, C., Kos, J., Erlingsson, Ü., & Song, D. (2018). The secret sharer: Measuring unintended neural network memorization & extracting secrets. *arXiv preprint arXiv:1802.08232*, 5.
- Chang, L., & Moskowitz, I. S. (1998). Parsimonious downgrading and decision trees applied to the inference problem. In Proceedings of the 1998 Workshop on New Security Paradigms. Chartottsville, VA. pp. 82–89.
- Chen, F., Luo, M., Dong, Z., Li, Z., & He, X. (2018). Federated meta-learning with fast convergence and efficient communication. *arXiv preprint arXiv:1802.07876*.
- Chen, M., Semiari, O., Saad, W., Liu, X., & Yin, C. (2019). Federated echo state learning for minimizing breaks in presence in wireless virtual reality networks. *IEEE Transactions on Wireless Communications*, 19(1), 177–191.
- Cheng, K., Fan, T., Jin, Y., Liu, Y., Chen, T., Papadopoulos, D., & Yang, Q. (2021). Secureboost: A lossless federated learning framework. *IEEE Intelligent Systems*.
- Choudhury, O., Gkoulalas-Divanis, A., Salonidis, T., Sylla, I., Park, Y., Hsu, G., & Das, A. (2019). Differential privacy-enabled federated learning for sensitive health data. *arXiv preprint arXiv:1910.02578*.
- Ciriani, V., Di Vimercati, S. D. C., Foresti, S., & Samarati, P. (2008). *K*-anonymous data mining: A survey. In *Privacy-preserving data mining* (pp. 105–136). Springer.
- Clifton, C., Kantarcioglu, M., Vaidya, J., Lin, X., & Zhu, M. Y. (2002). Tools for privacy preserving distributed data mining. *ACM Sigkdd Explorations Newsletter*, 4(2), 28–34.
- Doku, R., Rawat, D. B., & Liu, C. (2019). Towards federated learning approach to determine data relevance in big data. In 2019 IEEE 20th International Conference on Information Reuse and Integration for Data Science (IRI). Los Angeles, CA. pp. 184–192.
- Duan, M., Liu, D., Ji, X., Liu, R., Liang, L., Chen, X., & Tan, Y. (2020). Fedgroup: Ternary cosine similarity-based clustered federated learning framework toward high accuracy in heterogeneous data. *arXiv preprint arXiv:2010.06870*.
- Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). The KDD process for extracting useful knowledge from volumes of data. *Communications of the ACM*, 39(11), 27–34.
- Fu, A., Zhang, X., Xiong, N., Gao, Y., Wang, H., & Zhang, J. (2020). VFL: A verifiable federated learning with privacy-preserving for big data in industrial IoT. *IEEE Transactions on Industrial Informatics*. p. 1.
- Ganesh, S. H., & Christy, A. J. (2015). Applications of educational data mining: A survey. In 2015 International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS). Coimbatore, India. pp. 1–6.
- Geiping, J., Bauermeister, H., Droge, H., & Moeller, M. (2020). Inverting gradients—How easy is it to break privacy in federated learning? *arXiv preprint arXiv:2003.14053*.
- Geyer, R. C., Klein, T., & Nabi, M. (2017). Differentially private federated learning: A client level perspective. *arXiv preprint arXiv:1712.07557*.
- Glicksberg, B. S., Johnson, K. W., & Dudley, J. T. (2018). The next generation of precision medicine: Observational studies, electronic health records, biobanks and continuous monitoring. *Human Molecular Genetics*, 27(R1), R56–R62.
- Guo, S., & Zeng, D. (2020). Pedagogical data federation toward education 4.0. In Proceedings of the 2020 the 6th International Conference on Frontiers of Educational Technologies. Tokyo, Japan. pp. 51–55.
- Gupta, G. K. (2014). *Introduction to data mining with case studies*. PHI Learning Pvt. Ltd.
- Hsieh, K., Harlap, A., Vijaykumar, N., Konomis, D., Ganger, G. R., Gibbons, P. B., & Mutlu, O. (2017). Gaia: Geo-distributed machine learning approaching {LAN} speeds. In 14th {USENIX} Symposium on Networked Systems Design and Implementation (FNSDIG 17). Boston, MA. pp. 629–647.
- Huang, Y., Su, Y., Ravi, S., Song, Z., Arora, S., & Li, K. (2020). Privacy-preserving learning via deep net pruning. *arXiv preprint arXiv:2003.01876*.

- Hui-You, Z. F. C. (2006). Research on privacy-preserving collaborative filtering recommendation based on distributed data. *Chinese Journal of Computers*, 29(8), 237–245.
- Jiang, Y., Wang, S., Valls, V., Ko, B. J., Lee, W.-H., Leung, K. K., & Tassiulas, L. (2019). Model pruning enables efficient federated learning on edge devices. *arXiv preprint arXiv:1909.12326*.
- Kairouz, P., McMahan, H. B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A. N., Bonawitz, K. A., Charles, Z., Cormode, G., Cummings, R., D'Oliveira, R. G. L., Rouayheb, S. E., Evans, D., Gardner, J., Garrett, Z., Gascón, A., Ghazi, B., Gibbons, P. B., Gruteser, M. ... Zhao, S. (2019). Advances and open problems in federated learning. *arXiv preprint arXiv:1912.04977*.
- Kim, H., Park, J., Bennis, M., & Kim, S.-L. (2018). On-device federated learning via blockchain and its latency analysis. *arXiv preprint arXiv:1808.03949*.
- Konečný, J., McMahan, H. B., Yu, F. X., Richtárik, P., Suresh, A. T., & Bacon, D. (2016). Federated learning: Strategies for improving communication efficiency. *arXiv preprint arXiv:1610.05492*.
- Laguel, Y., Pillutla, K., Malick, J., & Harchaoui, Z. (2020). Device heterogeneity in federated learning: A superquantile approach. *arXiv preprint arXiv:2002.11223*.
- Larson, D. B., Magnus, D. C., Lungren, M. P., Shah, N. H., & Langlotz, C. P. (2020). Ethics of using and sharing clinical imaging data for artificial intelligence: A proposed framework. *Radiology*, 295(3), 675–682.
- Lee, J., Sun, J., Wang, F., Wang, S., Jun, C.-H., & Jiang, X. (2018). Privacy-preserving patient similarity learning in a federated environment: Development and analysis. *JMIR Medical Informatics*, 6(2), e7744.
- Li, L., Fan, Y., & Lin, K.-Y. (2020). A survey on federated learning. In 2020 IEEE 16th international conference on control & automation (ICCA). Singapore. pp. 791–796.
- Li, L., Fan, Y., Tse, M., & Lin, K.-Y. (2020). A review of applications in federated learning. *Computers & Industrial Engineering*, 149(5), 106854.
- Li, T., Sahu, A. K., Talwalkar, A., & Smith, V. (2020). Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine*, 37(3), 50–60.
- Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A., & Smith, V. (2018). Federated optimization in heterogeneous networks. *arXiv preprint arXiv:1812.06127*.
- Li, T., Sanjabi, M., Beirami, A., & Smith, V. (2019). Fair resource allocation in federated learning. *arXiv preprint arXiv:1905.10497*.
- Liang, P. P., Liu, T., Liu, Z., Allen, N. B., Auerbach, R. P., Brent, D., Salakhutdinov, R., & Morency, L.-P. (2020). Think locally, act globally: Federated learning with local and global representations. *arXiv preprint arXiv:2001.01523*.
- Liang, X., Liu, Y., Chen, T., Liu, M., & Yang, Q. (2019). Federated transfer reinforcement learning for autonomous driving. *arXiv preprint arXiv:1910.06001*.
- Liu, J., Yang, T., Bai, J., & Sun, B. (2021). Resource allocation and scheduling in the intelligent edge computing context. *Future Generation Computer Systems*, 121, 48–53.
- Liu, Y., James, J., Kang, J., Niyato, D., & Zhang, S. (2020). Privacy-preserving traffic flow prediction: A federated learning approach. *IEEE Internet of Things Journal*, 7(8), 7751–7763.
- Lu, Y., Huang, X., Dai, Y., Maharjan, S., & Zhang, Y. (2019a). Blockchain and federated learning for privacy-preserved data sharing in industrial IoT. *IEEE Transactions on Industrial Informatics*, 16(6), 4177–4186.
- Lu, Y., Huang, X., Dai, Y., Maharjan, S., & Zhang, Y. (2019b). Differentially private asynchronous federated learning for mobile edge computing in urban informatics. *IEEE Transactions on Industrial Informatics*, 16(3), 2134–2143.
- Luo, Y.-L., Huang, L.-S., Jing, W.-W., Yao, Y.-F., & Chen, G.-L. (2005). Algorithm for privacy-preserving Boolean association rule mining. *Dianzi Xuebao (Acta Electronica Sinica)*, 33(5), 900–903.
- Luping, W., Wei, W., & Bo, L. (2019). CMFL: Mitigating communication overhead for federated learning. In 2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS). Dallas, TX. pp. 954–964.
- Lyu, L., Yu, H., & Yang, Q. (2020). Threats to federated learning: A survey. *arXiv preprint arXiv:2003.02133*.
- Ma, C., Li, J., Ding, M., Yang, H. H., Shu, F., Quek, T. Q., & Poor, H. V. (2020). On safeguarding privacy and security in the framework of federated learning. *IEEE Network*, 34(4), 242–248.
- Malekijoo, A., Fadaeieslam, M. J., Malekijoo, H., Homayounfar, M., Alizadeh-Shabdiz, F., & Rawassizadeh, R. (2021). Fedzip: A compression framework for communication-efficient federated learning. *arXiv preprint arXiv:2102.01593*.
- Mammen, P. M. (2021). Federated learning: Opportunities and challenges. *arXiv preprint arXiv:2101.05428*.
- McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. In *Artificial Intelligence and Statistics*. pp. 1273–1282.
- Melis, L., Song, C., De Cristofaro, E., & Shmatikov, V. (2019). Exploiting unintended feature leakage in collaborative learning. In 2019 IEEE Symposium on Security and Privacy (SP). San Francisco, CA. pp. 691–706.
- Mo, F., & Haddadi, H. (2019). Efficient and private federated learning using TEE. In Proceedings of EuroSys Conference. Dresden, Germany.
- Mothukuri, V., Parizi, R. M., Pouriyeh, S., Huang, Y., Dehghantanha, A., & Srivastava, G. (2021). A survey on security and privacy of federated learning. *Future Generation Computer Systems*, 115, 619–640.
- Nguyen, D. C., Ding, M., Pathirana, P. N., Seneviratne, A., Li, J., & Poor, H. V. (2021). Federated learning for internet of things: A comprehensive survey. *arXiv preprint arXiv:2104.07914*.
- Nishio, T., & Yonetani, R. (2019). Client selection for federated learning with heterogeneous resources in mobile edge. In ICC 2019–2019 IEEE International Conference on Communications (ICC). Shanghai, China. pp. 1–7.

- Piech, C., Spencer, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L., & Sohl-Dickstein, J. (2015). Deep knowledge tracing. *arXiv preprint arXiv:1506.05908*.
- Qi, X., & Zong, M. (2012). An overview of privacy preserving data mining. *Procedia Environmental Sciences*, 12, 1341–1347.
- Qi, Y., Hossain, M. S., Nie, J., & Li, X. (2021). Privacy-preserving blockchain-based federated learning for traffic flow prediction. *Future Generation Computer Systems*, 117, 328–337.
- Samarakoon, S., Bennis, M., Saad, W., & Debbah, M. (2019). Distributed federated learning for ultra-reliable low-latency vehicular communications. *IEEE Transactions on Communications*, 68(2), 1146–1159.
- Savaglio, C., Gerace, P., Di Fatta, G., & Fortino, G. (2019). Data mining at the IoT edge. In 2019 28th International Conference on Computer Communication and Networks (ICCCN). Valencia, Spain. pp. 1–6.
- Shadroo, S., & Rahmani, A. M. (2018). Systematic survey of big data and data mining in internet of things. *Computer Networks*, 139, 19–47.
- Shah, M., & Nair, S. (2015). A survey of data mining clustering algorithms. *International Journal of Computer Applications*, 128(1), 1–5.
- Stripelis, D., & Ambite, J. L. (2021). Semi-synchronous federated learning. *arXiv preprint arXiv:2102.02849*.
- Sun, G., Cong, Y., Dong, J., Wang, Q., & Liu, J. (2020). Data poisoning attacks on federated machine learning. *arXiv preprint arXiv:2004.10020*.
- Sun, Z., Kairouz, P., Suresh, A. T., & McMahan, H. B. (2019). Can you really backdoor federated learning? *arXiv preprint arXiv:1911.07963*.
- Sweeney, L. (2002a). Achieving k-anonymity privacy protection using generalization and suppression. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(05), 571–588.
- Sweeney, L. (2002b). K-anonymity: A model for protecting privacy. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(05), 557–570.
- Tolpegin, V., Truex, S., Gursoy, M. E., & Liu, L. (2020). Data poisoning attacks against federated learning systems. In European Symposium on Research in Computer Security. Guildford, UK. pp. 480–501.
- Tomar, D., & Agarwal, S. (2013). A survey on data mining approaches for healthcare. *International Journal of Bio-Science and Bio-Technology*, 5(5), 241–266.
- Upreti, A., Rawat, D. B., & Li, J. (2021). Privacy preserving misbehavior detection in IoV using federated machine learning. In 2021 IEEE 18th Annual Consumer Communications & Networking Conference (CCNC). Las Vegas, NV. pp. 1–6.
- Vaghashia, H., & Ganatra, A. (2015). A survey: Privacy preservation techniques in data mining. *International Journal of Computer Applications*, 119(4), 20–26.
- Vaid, A., Jaladanki, S. K., Xu, J., Teng, S., Kumar, A., Lee, S., Somani, S., Paranjpe, I., De Freitas, J. K., Wanyan, T., Johnson, K. W., Bicak, M., Klang, E., Kwon, Y. J., Costa, A., Zhao, S., Miotto, R., Charney, A. W., Böttinger, E. ... Glicksberg, B. S. (2020). Federated learning of electronic health records improves mortality prediction in patients hospitalized with covid-19. *medRxiv*.
- Wang, X., Han, Y., Wang, C., Zhao, Q., Chen, X., & Chen, M. (2019). In-edge AI: Intelligentizing mobile edge computing, caching and communication by federated learning. *IEEE Network*, 33(5), 156–165.
- Wang, Z., Song, M., Zhang, Z., Song, Y., Wang, Q., & Qi, H. (2019). Beyond inferring class representatives: User-level privacy leakage from federated learning. In IEEE INFOCOM 2019-IEEE conference on computer communications. Paris, France. pp. 2512–2520.
- Weiping, G., Wei, W., Haofeng, Z., & Baile, S. (2006). Privacy preserving classification mining. *Journal of Computer Research and Development*, 43(1), 39.
- Wu, J., Huang, Z., Liu, Q., Lian, D., Wang, H., Chen, E., Ma, H., & Wang, S. (2021). Federated deep knowledge tracing. In Proceedings of the 14th ACM International Conference on web search and data mining. Virtual Event Israel. pp. 662–670.
- Wu, Q., He, K., & Chen, X. (2020). Personalized federated learning for intelligent IoT applications: A cloud-edge based framework. *IEEE Open Journal of the Computer Society*, 1, 35–44.
- Xie, M., Long, G., Shen, T., Zhou, T., Wang, X., & Jiang, J. (2020). Multi-center federated learning. *arXiv preprint arXiv:2005.01026*.
- Xu, W., Fang, W., Ding, Y., Zou, M., & Xiong, N. (2021). Accelerating federated learning for iot in big data analytics with pruning, quantization and selective updating. *IEEE Access*, 9, 38457–38466.
- Yao, A. C.-C. (1986). How to generate and exchange secrets. In 27th Annual Symposium on Foundations of Computer Science (SFCS 1986). Toronto, ON, Canada. pp. 162–167.
- Yao, H., Tang, X., Wei, H., Zheng, G., Yu, Y., & Li, Z. (2018). Modeling spatial-temporal dynamics for traffic prediction. *arXiv preprint arXiv:1803.01254*.
- Yu, H., Liu, Z., Liu, Y., Chen, T., Cong, M., Weng, X., Niyato, D., & Yang, Q. (2020a). A fairness-aware incentive scheme for federated learning. In Proceedings of the AAAI/ACM conference on AI, ethics, and society. New York, NY. pp. 393–399.
- Yuan, J., Xu, M., Ma, X., Zhou, A., Liu, X., & Wang, S. (2020). Hierarchical federated learning through LAN-WAN orchestration. *arXiv preprint arXiv:2010.11612*.
- Yurochkin, M., Agarwal, M., Ghosh, S., Greenewald, K., Hoang, N., & Khazaeni, Y. (2019). Bayesian nonparametric federated learning of neural networks. In International Conference on Machine Learning. Long Beach, CA. pp. 7252–7261.
- Zhang, C., Xie, Y., Bai, H., Yu, B., Li, W., & Gao, Y. (2021). A survey on federated learning. *Knowledge-Based Systems*, 216, 106775.
- Zhang, J., Chen, J., Wu, D., Chen, B., & Yu, S. (2019). Poisoning attack in federated learning using generative adversarial nets. In 2019 18th IEEE International Conference on Trust, Security and Privacy in Computing and Communications/13th IEEE International Conference on Big Data Science and Engineering (TrustCom/BigDataSE). Rotorua, New Zealand. pp. 374–380.

- Zhang, P., Tong, Y.-H., Tang, S.-W., Yang, D.-Q., & Ma, X.-L. (2006). An effective method for privacy preserving association rule mining. *Ruan Jian Xue Bao (Journal of Software)*, 17(8), 1764–1774.
- Zhang, W., Zhou, T., Lu, Q., Wang, X., Zhu, C., Sun, H., Wang, Z., Lo, S. K., & Wang, F.-Y. (2021). Dynamic fusion-based federated learning for covid-19 detection. *IEEE Internet of Things Journal*, 8, 15884–15891.
- Zhao, Y., Li, M., Lai, L., Suda, N., Civin, D., & Chandra, V. (2018). Federated learning with non-IID data. *arXiv preprint arXiv:1806.00582*.
- Zheng, Z., Zhou, Y., Sun, Y., Wang, Z., Liu, B., & Li, K. (2021). Applications of federated learning in smart cities: Recent advances, taxonomy, and open challenges. *Connection Science*, 1–28.
- Zhu, H., Li, Z., Cheah, M., & Goh, R. S. M. (2020). Privacy-preserving weighted federated learning within oracle-aided MPC framework. *arXiv preprint arXiv:2003.07630*.
- Zhuo, H. H., Feng, W., Lin, Y., Xu, Q., & Yang, Q. (2019). Federated deep reinforcement learning. *arXiv preprint arXiv:1901.08277*.

**How to cite this article:** Yu, B., Mao, W., Lv, Y., Zhang, C., & Xie, Y. (2022). A survey on federated learning in data mining. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 12(1), e1443. <https://doi.org/10.1002/widm.1443>